1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L38 – Concluding Remarks**

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3 **Our Goal**

- To design & implement human-computer interfaces (interactive systems)
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4 **Main Challenges**

- System should be
  - Usable
  - Executable

5 **Systematic Approach**

- Software development life cycle

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7 **Stages**

- Requirement stage – functional and usability (CI)
- Design– design guidelines (usability), DFD+ER, OOD+UML
- Prototype (including evaluation) – CW, HE
- Coding and code testing – black-box, white-box, integration + system testing
- Usability testing

8 **Stages**

- Requirement stage
  - Learned about functional requirement gathering and analysis
  - Learned about usability requirement gathering and analysis (CI)

9 **Stages**

- Design stage
- TWO designs
  - Interface design
  - Code/system design

10 **Stages**

- Interface design
- Design–prototype-evaluate cycle (to ensure usability)
  - Design guidelines and brain-storming
  - Prototype
  - Prototype evaluation – CW, HE

11 **Stages**

- Code/system design (to ensure efficient implementation)
  - Functional approach - DFD+ER,
  - Object-oriented approach - UML

12 **Stages**

- Implementation and code testing (to ensure execution-able code)
  - Unit-level testing – review-based, black-box, white-box
  - Integration + system testing

13 **Stages**

- Empirical study (to ensure usable end-product)

14 **How it Helps**

- Takes care of usability (very important)
- Takes care of efficient implementation (also very important)

15 **Outputs**

- Not only the end-product (software)
- Also, documentation (for maintenance)
  - SRS + usability requirement reports
  - Interface and code design documents (including prototype evaluation reports)
  - Code testing documents
  - Usability study reports
  - User & technical manuals
  - Help documents …

16 **Conclusion**

- The process and documentation important
  - Ensures *usable* & *executable* end product that is *maintainable* and also *acceptable* to users

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Thank you